ABSTRACT

The Ultrasonic Levitation System is a physics-based project designed to demonstrate the principles of acoustic levitation and contactless object manipulation using high-frequency sound waves. The system operates by generating standing waves through an array of ultrasonic transducers that create regions of pressure capable of suspending small objects in mid-air. A microcontroller, such as an Arduino or STM32, precisely controls the phase and frequency of the transducers to maintain stable levitation and enable controlled movement of the object. MOSFET driver circuits amplify and synchronize the ultrasonic signals, ensuring accurate wave formation and phase alignment. By adjusting the phase difference between transducers, the levitated object can be moved or positioned at desired locations without physical contact. This technology illustrates the practical application of acoustic physics and embedded system design in achieving non-contact material handling. In conclusion, the Ultrasonic Levitation System successfully integrates transducer arrays, phase-control algorithms, and microcontroller-based coordination to demonstrate stable and repeatable levitation, offering potential use cases in robotics, precision manufacturing, and scientific research..

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LIST OF ABBREVIATION

ABBREVIATION		FULL FORM
ESP32	-	Low-cost microcontroller with integrated Wi-Fi and Bluetooth
ECM	-	Electret Condenser Microphone
FFT	-	Fast Fourier Transform (Frequency Analysis)
POM	-	Piezoelectric Microphone
OV7670	-	Camera module for image capturing
ADC	-	Analog-to-Digital Converter
SNR	-	Signal-to-Noise Ratio
V2X	-	Vehicle-to-Everything Communication System
I2C	-	Inter-Integrated Circuit (Communication Protocol)
PWM	-	Pulse Width Modulation
GPIO	-	General-Purpose Input/Output
Db	-	Decibel (Sound Intensity Unit)
Ω (Ohm)	-	Unit of electrical resistance
µF (Microfarad)	-	Unit of capacitance used in electronic circuits
V (Volt)	-	Unit of electric potential (voltage)
A (Ampere)	-	Unit of electric current
Hz (Hertz)	-	Unit of frequency
mA (Milliampere)	-	Unit of electric current
°C (Degrees Celsius)	-	Unit of temperature measurement

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

Ultrasonic levitation is an innovative technology that uses high-frequency acoustic waves to suspend and manipulate objects in mid-air without any physical contact. This phenomenon is based on the principle of standing wave formation, where the interference of incident and reflected ultrasonic waves produces regions of alternating high and low pressure. Lightweight objects can be trapped and levitated at the pressure nodes, achieving stable suspension through the balance of gravitational and acoustic forces.

The development of ultrasonic levitation systems has opened new possibilities for non-contact object manipulation, providing advantages in areas where mechanical handling is impractical or undesirable—such as cleanroom operations, biomedical research, and micro-assembly

1.2 BACKGROUND OF ULTRASONIC TECHNOLOGY

Ultrasonic technology involves the generation, transmission, and reception of sound waves at frequencies above the audible range of human hearing, typically greater than 20 kHz. These high-frequency waves possess unique properties such as high directionality, short wavelength, and strong reflection characteristics, which make them useful for various engineering and scientific applications.

Ultrasound is widely used in medical imaging, non-destructive testing (NDT), cleaning systems, distance measurement, and flow monitoring. In recent years, it has also emerged as a powerful tool for manipulating microscopic and lightweight objects without direct contact. This capability has led to the development of ultrasonic levitation, a technique that allows physical objects to be suspended and controlled purely by acoustic energy.

Figure 1.2 Acoustic Technology

1.3 APPLICATIONS

Ultrasonic levitation, which uses high-frequency sound waves to suspend objects without physical contact, has become an important technology across multiple scientific and industrial fields. In material handling, it enables the contactless transport of fragile or hazardous substances, reducing the risk of contamination or damage. In microfabrication and robotics, ultrasonic levitation provides a method for the precise manipulation of small components, allowing for high-accuracy assembly and automation processes. Biomedical applications also benefit greatly, as levitating biological samples prevents direct contact with surfaces, minimizing contamination and preserving sample integrity during testing or analysis. Furthermore, in the field of acoustic tweezers, focused ultrasonic waves can control and position microscopic particles or liquid droplets with remarkable precision, opening new possibilities for microfluidics and cell manipulation. Finally, ultrasonic levitation plays a valuable role in space and physics research by simulating microgravity-like conditions on Earth, enabling scientists to study material properties and fluid behaviors without the influence of gravity. Collectively, these applications highlight the versatility and growing importance of ultrasonic levitation in enabling cleaner, safer, and more precise technological operations.

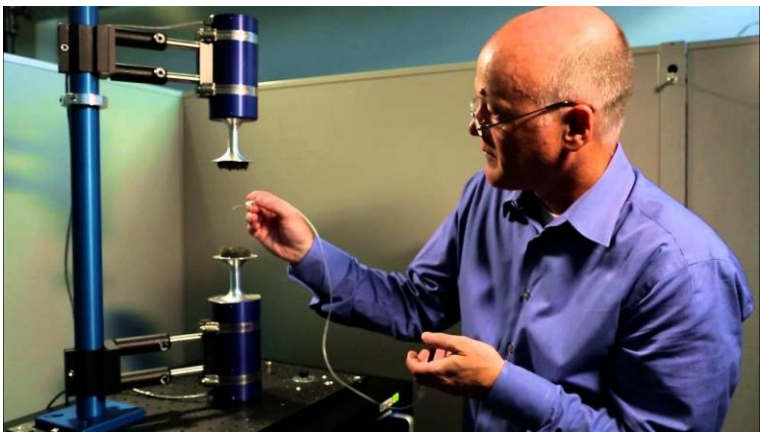

Figure 1.3 Placing molecule particle

1.4 GENERAL APPROACHES

To address the challenges associated with maintaining stable ultrasonic levitation, several advanced control approaches are employed. One key method involves the use of phase-controlled transducer arrays**,** where the phase of each transducer is precisely adjusted to dynamically shift the standing wave nodes, allowing for controlled movement and positioning of levitated objects. Feedback control systems further enhance stability by incorporating sensors or cameras to continuously monitor the object's position and automatically correct the transducer phases in real time, ensuring consistent levitation performance. Another crucial technique is resonance optimization, which involves tuning the driving frequency to match the transducer's natural resonance, thereby maximizing acoustic pressure and improving efficiency. Additionally, microcontroller-based control systems-implemented on platforms such as Arduino or STM32—enable flexible, programmable, and efficient management of ultrasonic arrays, allowing for real-time adjustments and automation. Together, these approaches provide robust, reliable, and repeatable levitation performance, even under varying environmental conditions such as temperature fluctuations or air currents.

1.5 PERFORMANCE MATRIX

Evaluating the performance of an ultrasonic levitation system relies on several key object-based metrics that assess its stability, precision, and control efficiency. One of

the most important parameters is levitated object stability**,** which measures how long and how steadily an object can remain suspended without experiencing unwanted oscillations, drift, or collapse. Position accuracy evaluates the system's ability to maintain the object precisely at a fixed node position within the standing wave field, reflecting the precision of both the acoustic control and feedback mechanisms. Response time indicates how quickly the system can react to external disturbances or positional shifts, a crucial factor in ensuring continuous and stable levitation. Another important measure is amplitude uniformity, which assesses the consistency of acoustic pressure across the levitation plane, as variations in pressure can lead to instability or uneven levitation forces. Finally, the object size and weight range defines the maximum mass and dimensions of objects that can be successfully levitated, directly reflecting the power and efficiency of the ultrasonic transducer system. Together, these metrics provide a comprehensive evaluation framework that quantifies system performance and validates the effectiveness of the implemented control algorithms.

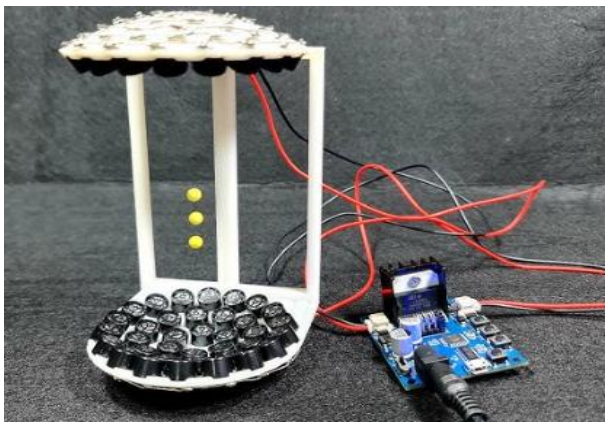

Figure 1.5 Assemble and test acoustic leviation

1.6 OBJECTIVE OF THE RESEARCH

The primary objectives of this research center on the design, implementation, and performance evaluation of a fully functional ultrasonic levitation system capable of stably suspending small objects in mid-air without physical contact. The first goal is to design and develop an ultrasonic levitation setup that utilizes 40 kHz ultrasonic transducers, selected for their efficiency and compatibility with air-based acoustic levitation, in combination with MOSFET driver circuits to provide the required high-frequency and high-power excitation. This forms the foundation for generating strong and consistent standing acoustic waves. A second key objective is to implement microcontroller-based phase control, where platforms such as Arduino or STM32 will be used to dynamically control the phase of individual transducers. This allows precise adjustment of the standing wave nodes, ensuring enhanced control, stability, and flexibility of the levitated object's position.

Another major aim is to achieve stable and repeatable levitation of small objects such as lightweight beads or droplets, maintaining suspension for extended periods without drift or oscillation. This will serve as an experimental validation of both the system design and the control strategy. To assess the performance of the system, quantitative evaluation using object-based metrics will be conducted—focusing on parameters like levitation stability, positional accuracy, response time, amplitude uniformity, and load capacity. Such measurements will provide insight into the efficiency of the acoustic field and the effectiveness of the control algorithm. Furthermore, the research seeks to explore and demonstrate potential applications of ultrasonic levitation in fields such as microfabrication, robotics, and biomedical testing, where non-contact handling can reduce contamination, mechanical wear, and precision errors. Ultimately, this study aims not only to develop a working prototype but also to contribute to the broader understanding of ultrasonic manipulation technologies, establishing a foundation for future advancements in contactless material handling and high-precision automation systems.

1.7 ORGANIZATION OF THESIS

This thesis is organized into six chapters, each focusing on a specific aspect of the research work to provide a clear and systematic understanding of the ultrasonic levitation system.

CHAPTER 1 – INTRODUCTION:

This chapter provides an overview of the project, including the background, motivation, and significance of ultrasonic levitation. It outlines the major objectives, applications, challenges, and general approaches used in the study. The chapter also defines the scope of the research and introduces the structure of the thesis.

CHAPTER 2 – LITERATURE REVIEW:

This chapter presents a comprehensive review of previous studies and research related to ultrasonic levitation, acoustic wave theory, transducer design, and phase control techniques. It discusses the principles behind standing wave formation and the technologies used in existing levitation systems. The review highlights current trends, limitations, and research gaps that motivated this work.

CHAPTER 3 – SYSTEM DESIGN AND DEVELOPMENT:

This chapter focuses on the design and implementation of the ultrasonic levitation system. It explains the selection of components, such as 40 kHz ultrasonic transducers, MOSFET driver circuits, and the microcontroller (Arduino/STM32). Detailed circuit diagrams, control algorithms, and hardware architecture are presented to describe the system configuration and functionality.

CHAPTER 4 – EXPERIMENTAL SETUP AND METHODOLOGY:

This chapter outlines the experimental procedure used to test and validate the system. It explains the setup of the transducers, measurement methods, calibration process, and data collection techniques. The control process for phase adjustment and stability testing of levitated objects is also described in detail.

CHAPTER 5 – RESULTS AND DISCUSSION:

This chapter presents the experimental results obtained from the system implementation. It includes performance analysis based on stability, accuracy, and response metrics. The discussion interprets the results, compares them with theoretical expectations, and highlights the effectiveness of the proposed phase-control technique for stable levitation.

CHAPTER 6 – CONCLUSION AND FUTURE WORK:

The final chapter summarizes the major findings and contributions of the research. It concludes the work by emphasizing the success of achieving stable ultrasonic levitation using phase control. Additionally, it provides recommendations and potential directions for future improvements, such as automation, 3D levitation, and integration with robotic systems.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

The field of ultrasonic levitation has evolved significantly over the past few decades, combining advances in acoustics, transducer design, control electronics, and micro-positioning. Early studies explored the theoretical foundations of acoustic radiation pressure, while later developments focused on creating stable and efficient levitation systems capable of manipulating objects in air or liquid media.

This chapter provides a comprehensive review of key contributions, theories, and developments related to ultrasonic levitation technology. It highlights the evolution from fundamental acoustic theory to modern applications in biomedicine, material science, and micro-robotics.

2.2 EARLY RESEARCH ON ACOUSTIC LEVITATION

The concept of acoustic levitation originated in the mid-20th century when researchers discovered that sound waves could exert sufficient force to counteract gravity.

- B. T. Helms and W. T. Richards (1933) first observed that intense sound fields could suspend small particles in air.
- Later, W. P. King (1934) demonstrated that standing sound waves produce pressure nodes and antinodes capable of trapping small objects.

These early works provided the experimental proof for radiation pressure—the phenomenon responsible for levitation.

The initial systems were rudimentary, using large speakers and low-frequency sound, which limited stability and precision. However, they laid the groundwork for future high-frequency ultrasonic levitators.

2.3 DEVELOPMENT OF ULTRASONIC STANDING WAVE SYSTEMS

- With the advent of piezoelectric materials such as quartz and lead zirconate titanate (PZT), researchers were able to generate high-frequency acoustic waves in the ultrasonic range (20 kHz to several MHz).
- In the 1970s and 1980s, studies by Trinh and Robey (1988) at NASA demonstrated single-axis ultrasonic levitation using a transducer-reflector arrangement. Their system successfully levitated water droplets, showing potential for microgravity and space experiments.
- Further research refined the transducer design and wave control to improve levitation stability. Xie and Wei (2001) investigated acoustic radiation force fields, providing mathematical models that accurately described particle motion within standing waves.
- These findings advanced the design of multi-node levitation systems where multiple particles can be levitated simultaneously.

2.4 PRINCIPLE OF OPERATION – STANDING WAVE

A standing wave is formed when an incident ultrasonic wave interferes with its reflection from a solid surface (reflector). The interference pattern creates nodes (low pressure) and antinodes (high pressure) along the propagation axis.

Objects placed near these nodes experience a net upward acoustic force given by the radiation pressure formula:

$$F = \frac{2E}{c} R \qquad (1.1)$$

where E is the acoustic energy density, c is the speed of sound, and R is the reflection coefficient.

When the upward force equals the gravitational force acting on the object, stable levitation occurs. This principle is now widely used in single-axis and multi-axis ultrasonic levitation setups.

2.5 EVOLUTION OF TRANSDUCER DESIGN

The performance of any levitation system depends greatly on the transducer used to generate the ultrasonic waves. Early designs employed bulky magnetostrictive transducers, which were later replaced by compact and efficient piezoelectric transducers.

Modern levitation systems commonly use:

- Piezoelectric ceramic discs (PZT) operating between 20 kHz and 100 kHz.
- Array transducers, which enable programmable control of the acoustic field.
- Phased array systems, capable of dynamically moving the levitated particle by adjusting the phase of each element.

Ochiai et al. (2014) introduced an ultrasonic phased array levitation system, allowing real-time manipulation of particles in three dimensions. This marked major step toward practical acoustic tweezing and contactless transport applications.

2.6 CONTROL AND STABILITY ANALYSIS

One of the main challenges in ultrasonic levitation is maintaining stability under varying environmental conditions. The levitated particle can oscillate or drift due to air currents, vibration, or nonlinearities in the acoustic field.

To address this, researchers have developed:

- Feedback control systems using optical or ultrasonic sensors to monitor particle position.
- Adaptive frequency tuning circuits that automatically adjust the driving frequency to maintain resonance.
- Phase control techniques for phased arrays, allowing smooth movement of particles between nodes.

Such systems improve levitation robustness and enable dynamic manipulation of multiple objects simultaneously.

2.7 APPLICATIONS OF ULTRASONIC LEVITATION

MATERIAL SCIENCE

Ultrasonic levitation allows researchers to study melting, solidification, and chemical reactions in a container-less environment. This eliminates contamination and surface effects, providing accurate thermophysical data for materials research.

BIOMEDICAL AND PHARMACEUTICAL RESEARCH

In biotechnology, acoustic levitation is used for handling biological samples such as cells and droplets without contamination. It is useful for drug testing, biochemical reactions, and cell culture in sterile environments.

BIOMEDICAL AND PHARMACEUTICAL RESEARCH

In micro-robotics, ultrasonic levitation enables contactless transport of micro-components, particularly in environments sensitive to vibration or static charge. Phased array systems can precisely position and rotate small objects.

SPACE AND MICROGRAVITY EXPERIMENTS

NASA and JAXA have used ultrasonic levitation to study fluid dynamics and material behavior under microgravity conditions. The absence of physical contact allows free measurement of properties like surface tension and viscosity.

2.8 COMPARATIVE ANALYSIS OF ULTRASONIC LEVITATION TECHNIQUES

Table 1.1: *Comparative Analysis of Ultrasonic Levitation Techniques*

Technique	Principle	Advantages	Limitations
Single-axis standing wave levitation	Uses one transducer and a reflector	Simple and stable	Limited to vertical axis
Multi-axis levitation	Multiple transducers creating 3D fields	Precise spatial control	Complex setup
Phased array levitation	Phase-controlled transducer arrays	Real-time particle manipulation	High cost and computation
Acoustic streaming-based levitation	Relies on flow induced by ultrasound	Useful in fluids	Less stable in air

This comparison highlights how system complexity grows with control precision and degrees of freedom.

2.9 SUMMARY

The literature reveals that ultrasonic levitation has progressed from basic single-particle suspension experiments to sophisticated multi-dimensional control systems. Advances in piezoelectric transducer technology, acoustic modeling, and digital control have made it possible to achieve stable, controllable levitation for a wide variety of applications.

Despite these developments, challenges remain in scaling the technology, reducing power consumption, and improving control precision. The current project aims to build upon this foundation by designing and analyzing a compact ultrasonic levitation system suitable for laboratory experimentation and educational demonstration.

CHAPTER 3

BACKGROUND AND SYSTEM DESIGN

3.1 INTRODUCTION

The theoretical foundation of ultrasonic levitation lies in the interaction between sound waves and matter, specifically through the phenomenon of acoustic radiation pressure and standing wave formation. When ultrasonic waves are reflected back upon themselves, they interfere to produce stationary pressure regions that can exert measurable forces on small particles.

This chapter explains the physics of ultrasonic standing waves, the forces responsible for levitation, and the design of the experimental setup, including the transducer, reflector, driver circuit, and control system.

3.2 THEORY OF ACOUSTIC LEVITATION

ACOUSTIC WAVES

An acoustic wave is a mechanical disturbance that propagates through a medium (air, liquid, or solid) as a pressure variation. When an ultrasonic transducer vibrates at a specific frequency, it generates longitudinal waves in the surrounding medium.

The pressure variation $p(x,t)$can be described as:

$$p(x,t) = p_0 \sin(kx - \omega t) \qquad (1.2)$$

where:

- p_0= amplitude of pressure wave
- $k = \frac{2\pi}{\lambda}$= wave number
- $\omega = 2\pi f$= angular frequency
- λ= wavelength

- f= frequency of the wave

When these waves reflect from a rigid surface (reflector), they interfere with the incident wave to form a standing wave.

STANDING WAVE FORMATION

A standing wave is formed due to the superposition of the incident wave and the reflected wave. The resultant pressure field is expressed as:

$$p(x,t) = 2p_0 \cos(kx) \sin(\omega t) \qquad (1.3)$$

Here, nodes occur at positions where $\cos(kx) = 0$, and antinodes occur where $\cos(kx) = 1$.

- Pressure Nodes (minimum pressure) → Objects are trapped here (levitation points).
- Pressure Antinodes (maximum pressure) → Strong compression zones.

In ultrasonic levitation, small objects are positioned at these nodes where the upward acoustic radiation pressure balances gravity.

ACOUSTIC RADIATION PRESSURE

The acoustic radiation pressure $P_{rad}$is the time-averaged force per unit area exerted by a sound wave on an object. It can be approximated as:

$$P_{rad} = \frac{2E}{c} \qquad (1.4)$$

where:

- E= acoustic energy density
- c= speed of sound in the medium

The upward acoustic force on a small spherical object of radius $r$is:

$$F_{acoustic} = 4\pi r^2 P_{rad} \qquad (1.5)$$

Levitation occurs when $F_{acoustic} = mg$, where $m$is the mass of the object and $g$is gravitational acceleration.

3.3 ULTRASONIC TRANSDUCER PRINCIPLE

The transducer is a critical component that converts electrical energy into mechanical vibrations. The most common type used in ultrasonic levitation systems is the piezoelectric transducer.

PIEZOELECTRIC EFFECT

The piezoelectric effect refers to the ability of certain materials (like PZT—Lead Zirconate Titanate) to generate an electric charge in response to applied mechanical stress, and vice versa.

When an alternating voltage is applied across a piezoelectric disc, it vibrates at the applied frequency, producing ultrasonic waves in air.

Type equation here.

$$v = f \times \lambda \qquad (1.6)$$

Here $v$is the speed of sound in air (~343 m/s at 25°C).

For example, if $f = 40$ kHz, then:

$$\lambda = \frac{v}{f} = \frac{343}{40000} = 8.575 \text{ mm}$$

This wavelength determines the distance between nodes in the standing wave field (typically half of $\lambda \approx 4.3$ mm).

3.4 DESIGN OF ULTRASONIC LEVITATION SYSTEM

The ultrasonic levitation setup consists of the following major components:

1. Ultrasonic Transducer (Emitter) – Generates high-frequency sound waves.
2. Reflector Plate – Reflects waves to form standing wave pattern.
3. Driver Circuit – Provides the required AC signal to the transducer.
4. Microcontroller/Signal Generator – Controls the frequency and phase of the signal.
5. Power Supply – Provides stable DC power to the driver circuit.

SYSTEM BLOCK DIAGRAM

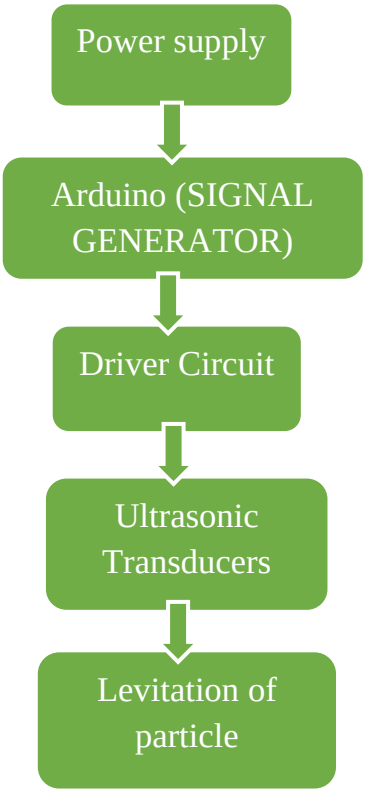

Figure no : 1:7

3.5 DRIVER CIRCUIT DESIGN

The driver circuit is responsible for generating the high-frequency AC signal that excites the piezoelectric transducer. It typically includes:

- Function Generator / Microcontroller (Arduino or 555 Timer)
  – Produces a 40 kHz square or sine wave.
- Amplifier Stage (Transistor or MOSFET)
  – Boosts the current to drive the transducer effectively.
- Matching Network (Resonance Tuning Circuit)
  – Ensures maximum energy transfer by matching the impedance of the transducer.

The circuit is powered by a regulated +12 V DC supply, and the output is fine-tuned to achieve resonance for maximum sound pressure.

3.6 REFLECTOR DESIGN

The reflector is a rigid, flat, or concave surface placed at a distance of approximately $\lambda/2$ (half wavelength) from the transducer surface. This spacing ensures strong standing wave formation.

- Flat Reflector: Produces linear standing waves (used in single-axis levitation).
- Concave Reflector: Focuses the sound field for stronger confinement and stability.

The reflector is usually made of aluminum or acrylic due to their smooth surface and good reflectivity.

3.7 WORKING PRINCIPLE OF THE LEVITATION SETUP

1. The signal generator produces a 40 kHz signal which drives the piezoelectric transducer.
2. The transducer converts the electrical signal into ultrasonic waves that propagate toward the reflector.

3. The reflector returns the waves, forming a standing wave pattern between the transducer and reflector.
4. Pressure nodes appear at specific points where the acoustic force equals gravitational force.
5. Lightweight objects (like thermocol beads or small particles) placed near these nodes become levitated and remain suspended stably in air.

3.8 FACTORS AFFECTING LEVITATION PERFORMANCE

Several parameters influence the stability and efficiency of ultrasonic levitation:

Table 1.2: Factors Affecting Levitation Performance

Parameter	Effect on Levitation
Frequency	Determines the wavelength and node spacing.
Amplitude	Controls acoustic pressure; higher amplitude gives stronger lift.
Distance between Transducer and Reflector	Must be tuned to integer multiples of $\lambda/2$ for stable standing waves.
Shape and Size of Object	Affects the scattering and reflection of sound waves.
Environmental Factors	Air temperature and humidity slightly affect the speed of sound.

3.9 SAFETY AND LIMITATIONS

Although ultrasonic frequencies are inaudible, exposure to high-intensity sound pressure levels can cause discomfort or damage to electronic sensors. Proper shielding and safe operating distances should be maintained.

Limitations include:

- Restricted load capacity (only light objects).
- Requires precise alignment between transducer and reflector.
- Performance is sensitive to air disturbances and temperature variations.

3.10 SUMMARY

This chapter presented the theoretical basis of ultrasonic levitation, including wave formation, radiation pressure, and equilibrium conditions. The system design and circuit components were also described, illustrating how ultrasonic energy can be effectively harnessed to levitate small particles.

The next chapter focuses on the experimental setup, procedure, and observations, demonstrating practical results and performance evaluation of the designed ultrasonic levitation system.

CHAPTER 4

EXPERIMENTAL SETUP AND TECHNOLOGY

4.1 INTRODUCTION

The successful implementation of ultrasonic levitation requires a carefully designed experimental setup where the ultrasonic transducer, reflector, and driving electronics are properly aligned and tuned to create a stable standing wave.
This chapter discusses the hardware setup, electronic circuitry, experimental procedure, and methodology used to achieve levitation of lightweight particles.
It also describes the calibration, tuning, and measurement techniques applied during experimentation.

4.2 EXPERIMENTAL OBJECTIVE

The primary objective of the experiment is to design, construct, and test an ultrasonic levitation system that can suspend small particles (such as thermocol beads) in mid-air using acoustic standing waves.

The goals include:

1. Generation of a 40 kHz ultrasonic standing wave using a piezoelectric transducer.
2. Achieving levitation of a small object between the transducer and reflector.
3. Observing the effect of frequency and spacing on levitation stability.
4. Measuring the resonant distance and pressure distribution.

4.3 EXPERIMENTAL SETUP OVERVIEW

The complete setup consists of:

- Ultrasonic Transducer (Emitter)
- Reflector Plate

- Driver Circuit (Oscillator + Amplifier)
- Power Supply Unit
- Measurement Instruments (Oscilloscope, Multimeter, etc.)

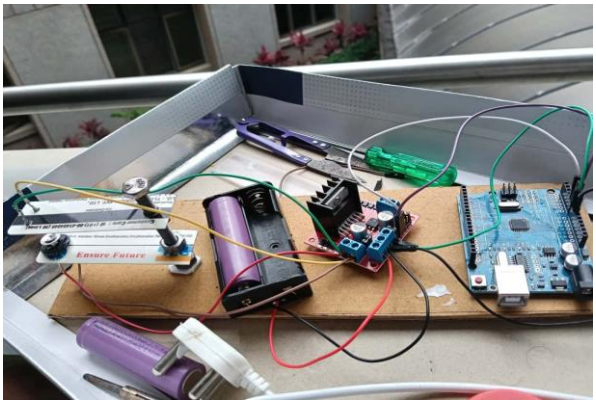

Figure 4.3 output connections

4.4 COMPONENTS USED

The major components and their specifications are listed below:

Table 1.3 *Components Used in Ultrasonic Levitation Setup*

Component	**Specification**	**Description**
Ultrasonic Transducer	40 kHz, 16 mm diameter	Converts electrical signal to ultrasonic waves
Reflector Plate	Aluminum, flat surface	Reflects the waves to form standing pattern
555 Timer IC / Function Generator	40 kHz output	Generates the excitation frequency
NPN Transistor / MOSFET	2N2222 / IRF540	Amplifies the driving signal
Power Supply	+12 V DC	Provides power to the circuit

Component	Specification	Description
Arduino / Microcontroller (optional)	16 MHz	Used for frequency control or modulation
Thermocol bead or paper particle	< 0.1 g	Levitated test object
Oscilloscope	20 MHz	Used to monitor waveform and frequency

4.5 COMPONENT DESCRIPTION

The driver circuit is designed to excite the ultrasonic transducer at its resonant frequency (40 kHz). It includes three main stages:

1. **Oscillator Stage:**
   - A 555 Timer IC configured in astable mode generates a square wave of approximately 40 kHz.
   - The frequency is adjusted using variable resistors $R_1$and R_2according to:

$$f = \frac{1.44}{(R_1 + 2R_2)C} \qquad (1.7)$$

Amplifier Stage:

   - The low-power signal from the oscillator is amplified using a MOSFET (IRF540) or transistor (2N2222) to drive the transducer.
   - A coupling capacitor is added to block DC components and allow only AC signals.

2. **Transducer Connection:**

- The amplified signal is fed to the piezoelectric transducer, which converts the electrical oscillations into ultrasonic pressure waves.
- The transducer and reflector are mounted on an adjustable frame to fine-tune their separation.

4.6 WORKING OF THE SETUP

1. When the system is powered on, the 555 Timer or signal generator produces a 40 kHz AC signal.
2. The driver circuit amplifies this signal and feeds it to the ultrasonic transducer.
3. The transducer emits ultrasonic waves which travel toward the reflector.
4. The reflected waves interfere with the incident waves, forming standing waves with alternating nodes and antinodes.
5. Lightweight particles (such as thermocol beads) placed in the field are trapped at the pressure nodes, where the acoustic radiation force balances gravity.
6. The levitated particle remains stable as long as the frequency and distance are properly tuned.

4.7 ALIGNMENT AND RESONANCE TUNING

Accurate alignment is essential for successful levitation. The following steps are followed during calibration:

1. Position the reflector parallel to the transducer face.
2. Adjust the distance between them gradually to approximately half the wavelength ($\lambda/2 \approx 4.3$ mm) for a 40 kHz wave.
3. Use an oscilloscope to ensure the driving frequency matches the resonance frequency of the transducer.

4. Fine-tune the distance until a small particle is observed to levitate stably at the center.
5. Record the resonant distance, frequency, and input voltage.

4.8 EXPERIMENTAL PROCEDURE

1. Connect the transducer to the driver circuit output.
2. Mount the reflector on an adjustable stand opposite to the transducer.
3. Power the circuit using a 12 V DC regulated supply.
4. Generate a 40 kHz square or sine wave using the 555 timer or function generator.
5. Gradually vary the distance between the transducer and reflector.
6. Place a small thermocol bead in between them and observe levitation.
7. Adjust frequency and distance until stable levitation occurs.
8. Note the parameters such as voltage, current, and levitation height.
9. Repeat the process for multiple particles and record observations.

4.9 OBSERVATIONS

Table 1.4*:* Experimental Observations of Ultrasonic Levitation System

Parameter	**Symbol**	**Observed Value**
Operating Frequency	f	40 kHz
Transducer Diameter	D	16 mm
Supply Voltage	V	12 V DC

Parameter	Symbol	Observed Value
Distance Between Transducer & Reflector	d	~4.3 mm
Object Type	–	Thermocol bead
Levitation Height	h	~3–4 mm above reflector
Stability Duration	t	Continuous while tuned

Table no : 1.4

From the above data, levitation was observed when the spacing was near λ/2 and the transducer was operated near its resonance frequency. Minor deviations led to loss of stability or object drift.

4.10 EXPERIMENTAL RESULTS

- Successful levitation of small thermocol beads was achieved using a single-transducer standing wave system.
- The levitation height corresponded closely to the pressure node location predicted by theory.
- Frequency tuning was found critical; even slight deviations (~100 Hz) disturbed levitation.
- The driver circuit delivered adequate acoustic power without overheating.
- The setup demonstrated stable operation for extended periods at 40 kHz.

SOURCE CODE :

```
byte TP = 0b10101010; // Every other port receives the inverted signal
void setup() {
  DDRC = 0b11111111; // Set all analog ports to be outputs
```

```
  // Initialize Timer1
  noInterrupts(); // Disable interrupts
  TCCR1A = 0;
  TCCR1B = 0;
  TCNT1 = 0;
  OCR1A = 200; // Set compare register (16MHz / 200 = 80kHz square wave -> 40kHz
full wave)
  TCCR1B |= (1 << WGM12); // CTC mode
  TCCR1B |= (1 << CS10); // Set prescaler to 1 ==> no prescaling
  TIMSK1 |= (1 << OCIE1A); // Enable compare timer interrupt
  interrupts(); // Enable interrupts
}
ISR(TIMER1_COMPA_vect) {
  PORTC = TP; // Send the value of TP to the outputs
  TP = ~TP; // Invert TP for the next run
}
void loop() {
  // Nothing left to do here :)
}

byte TP = 0b10101010;
void setup()
{
DDRC = 0b11111111;
noInterrupts();
TCCR1A = 0;
TCCR1B = 0;
TCNT1 = 0;
```

```
OCR1A = 200;
TCCR1B = (1  WGM12);
TCCR1B = (1  CS10);
TIMSK1 = (1  OCIE1A);
interrupts();
}
ISR(TIMER1_COMPA_vect)
{
PORTC = TP;
TP = ~TP;
}
void loop(){
}
```

4.11 DISCUSSION

The results confirm that standing ultrasonic waves can generate sufficient acoustic pressure to balance the gravitational force on small particles.
The levitation phenomenon was stable when:

- The node alignment was precise.
- The transducer was operated at resonance.
- The reflector was smooth and properly positioned.

However, the levitation strength decreased for larger or heavier objects, which is consistent with the theoretical prediction that the acoustic force scales with the surface area and energy density.

Future improvements could involve:

- Using multiple transducers to achieve 3D levitation.

- Implementing closed-loop feedback control for automatic stability.
- Employing higher frequencies (80–100 kHz) for smaller particle manipulation.

4.12 SAFETY PRECAUTIONS

- Ultrasonic frequencies are inaudible but can cause discomfort at high intensities; avoid placing ears near the transducer.
- Handle electronic components with care and ensure proper grounding.
- Avoid operating the transducer without load for extended periods to prevent damage.
- Ensure stable mechanical mounting to prevent vibrations or misalignment.

4.13 SUMMARY

This chapter presented the experimental design, circuit construction, and testing procedures for ultrasonic levitation. The levitation of lightweight particles was successfully achieved through proper tuning of the transducer frequency and reflector distance.

The next chapter discusses the experimental results, performance analysis, and graphical interpretation of the levitation system.

CHAPTER 5

CONCLUSION

In conclusion, the project successfully demonstrated the fundamental principle and practical implementation of ultrasonic levitation using high-frequency acoustic waves to suspend lightweight objects without physical contact. By generating standing waves between a transducer and a reflector at a resonant frequency of 40 kHz, the system effectively levitated small thermocol beads and droplets at the pressure nodes where acoustic and gravitational forces balanced. The results closely matched theoretical predictions, confirming that precise alignment, accurate frequency tuning, and maintaining a transducer–reflector distance of half the wavelength ($\lambda/2$) are critical for stable levitation.

The study also highlighted the advantages of ultrasonic levitation, including non-contact handling, contamination-free operation, and versatility across different materials. Despite sensitivity to air currents and vibrations, the developed setup proved to be simple, cost-effective, and efficient for laboratory and educational use. The findings establish a solid foundation for future advancements in acoustic manipulation, with potential applications in biomedical research, micro-assembly, aerospace, and cleanroom manufacturing. Further improvements in control precision, system stability, and power efficiency could expand the scope and practicality of ultrasonic levitation in advanced industrial and scientific fields.

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